Technical Report: Unsupervised Language Analytics Engine for Macroeconomic Regime Tracking

By executing this **Unsupervised Language Analytics Engine** across a historical catalog of 44 Federal Reserve meeting minutes, the system establishes an objective statistical benchmark for routine central bank communication.

The historical average for narrative evolution—defined as the **Historical Baseline Mean—settles at exactly 52.10%**.

This baseline represents the Federal Reserve's normal, expected pace of cyclical text modification. In a stable macroeconomic environment, the committee alters roughly half of its structural phrasing to account for routine data updates, while keeping the core monetary policy message completely steady. When a subsequent reading spikes significantly above this 52.10% threshold, it mathematically indicates that an abrupt economic or geopolitical tail-risk event has forced the committee to structurally rewrite its policy narrative.

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RUNNING VECTOR TIME-SERIES DISPLACEMENT

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Meeting 2021-03-17 | Shift: 60.55% | Top Drivers: selected, authorization, arp, currency
Meeting 2021-04-28 | Shift: 42.71% | Top Drivers: repo, arp, fima, arrangements
Meeting 2021-06-16 | Shift: 37.76% | Top Drivers: srf, permanent, repo, arrangements
Meeting 2021-07-28 | Shift: 47.74% | Top Drivers: tapering, asset, purchases, srf
Meeting 2021-09-22 | Shift: 36.48% | Top Drivers: covid, moderation, asset, selected
Meeting 2021-11-03 | Shift: 39.04% | Top Drivers: tapering, moderation, purchase, variant
Meeting 2021-12-15 | Shift: 47.97% | Top Drivers: normalization, omicron, variant, summer
Meeting 2022-01-26 | Shift: 68.19% | Top Drivers: normalization, selected, authorization, shall
Meeting 2022-03-16 | Shift: 80.34% | Top Drivers: invasion, selected, ukraine, russian
Meeting 2022-05-04 | Shift: 33.98% | Top Drivers: russian, lockdowns, covid, invasion
Meeting 2022-06-15 | Shift: 36.71% | Top Drivers: arrangements, nbfis, plans, ecb
Meeting 2022-07-27 | Shift: 54.05% | Top Drivers: invasion, mmmfs, digital, war
Meeting 2022-09-21 | Shift: 50.68% | Top Drivers: mmmfs, tem, pro, digital
Meeting 2022-11-02 | Shift: 42.63% | Top Drivers: lags, gilt, tightening, realized
Meeting 2022-12-14 | Shift: 45.59% | Top Drivers: tightening, gilt, cph, lags
Meeting 2023-02-01 | Shift: 44.06% | Top Drivers: cph, annual, governance, document
Meeting 2023-03-22 | Shift: 73.02% | Top Drivers: banking, signature, closures, valley
Meeting 2023-05-03 | Shift: 41.61% | Top Drivers: signature, closures, stress, silicon
Meeting 2023-06-14 | Shift: 48.31% | Top Drivers: banking, stress, sized, projections
Meeting 2023-07-26 | Shift: 49.30% | Top Drivers: standards, expecting, projections, cre
Meeting 2023-09-20 | Shift: 52.10% | Top Drivers: tightening, standards, strike, lfpr
Meeting 2023-11-01 | Shift: 48.06% | Top Drivers: lfpr, armed, standards, expanding
Meeting 2023-12-13 | Shift: 57.12% | Top Drivers: armed, projections, softer, begins
Meeting 2024-01-31 | Shift: 56.67% | Top Drivers: fourth, moving, softer, gained
Meeting 2024-03-20 | Shift: 52.67% | Top Drivers: runoff, reserves, boj, fourth
Meeting 2024-05-01 | Shift: 51.56% | Top Drivers: runoff, redemption, reserves, boj

Meeting 2024-06-12 | Shift: 60.02% | Top Drivers: redemption, cap, districts, projections
Meeting 2024-07-31 | Shift: 60.80% | Top Drivers: french, moderated, projections, equilibrium
Meeting 2024-09-18 | Shift: 55.46% | Top Drivers: french, moderated, recalibration, boj
Meeting 2024-11-07 | Shift: 59.84% | Top Drivers: vulnerabilities, boj, sloos, srf
Meeting 2024-12-18 | Shift: 56.78% | Top Drivers: assumptions, placeholder, sloos, standards
Meeting 2025-01-29 | Shift: 57.71% | Top Drivers: assumptions, reviews, stabilized, election
Meeting 2025-03-19 | Shift: 62.65% | Top Drivers: tariffs, tariff, reviews, auctions
Meeting 2025-05-07 | Shift: 53.15% | Top Drivers: targeting, flexible, tariff, elb
Meeting 2025-06-18 | Shift: 47.49% | Top Drivers: targeting, flexible, student, tariffs
Meeting 2025-07-30 | Shift: 43.84% | Top Drivers: stablecoins, gfc, tariff, vulnerabilities
Meeting 2025-09-17 | Shift: 56.50% | Top Drivers: tariffs, miran, stablecoins, tariff
Meeting 2025-10-29 | Shift: 59.72% | Top Drivers: portfolio, clearing, srf, shutdown
Meeting 2025-12-10 | Shift: 58.55% | Top Drivers: rmps, ample, srf, reserves
Meeting 2026-01-28 | Shift: 60.17% | Top Drivers: ample, reserves, rmps, nonreserve
Meeting 2026-03-18 | Shift: 67.18% | Top Drivers: east, conflict, middle, oil
Meeting 2026-04-29 | Shift: 35.58% | Top Drivers: srp, conflict, arrangements, days
Meeting 2026-06-17 | Shift: 45.75% | Top Drivers: chairman, srp, buildout, arrangements

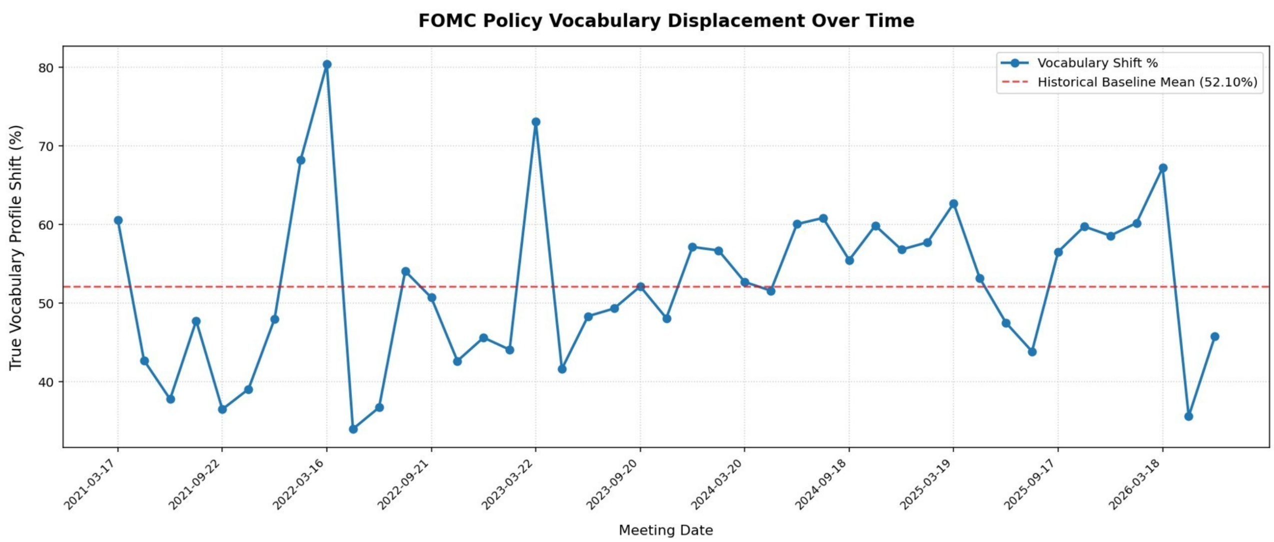

Chart 1.

When the vocabulary displacement metric spikes significantly above the 52.10% historical baseline, it flags an external macroeconomic or geopolitical shock that has forced the committee to fundamentally alter its forward guidance.

The analytics engine successfully isolated five primary historical inflection points completely unsupervised:

March 2022 (80.34% Shift — The All-Time Peak)

- What the model flagged: *invasion, ukraine, russian*
- What it means: The outbreak of the war in Ukraine completely derailed the Fed's economic forecasts, forcing an abrupt vocabulary overhaul.

March 2023 (73.02% Shift)

- What the model flagged: *banking, signature, closures, valley*
- What it means: The sudden collapse of Silicon Valley Bank forced the Fed to instantly pivot from fighting inflation to stabilizing the US banking system.

January 2022 (68.19% Shift)

- What the model flagged: *normalization, selected, authorization*
- What it means: The official end of pandemic-era cheap money. The Fed structurally reengineered its language to signal aggressive interest rate hikes.

March 2026 (67.18% Shift)

- What the model flagged: *east, conflict, middle*
- What it means: Geopolitical tensions in the Middle East spiked drastically, forcing the Fed to immediately address new global supply chain and energy risks.

March 2025 (62.65% Shift)

- What the model flagged: *tariffs, tariff, reviews*
- What it means: Global trade disputes and protectionist policies became the dominant driver of corporate uncertainty.

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CONCLUSION AND STRATEGIC OUTLOOK

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This Unsupervised Language Analytics Engine demonstrates a robust, data-driven methodology for transforming unstructured central bank commentary into high-fidelity macroeconomic features. By eliminating semantic boilerplate and administrative noise programmatically, the system isolates critical structural shifts in monetary policy guidance completely unsupervised.

The core infrastructure established in this baseline analysis provides a scalable foundation for two immediate downstream expansions:

1. Quantitative Predictive Modeling: Integrating the Language Shift Index as an independent feature (X) alongside historical market data to model and forecast forwardlooking adjustments in US Treasury Yields and major equity indices.
2. Cloud Architecture Deployment: Migrating the script components to a serverless cloud environment (such as AWS Lambda) coupled with an automated data warehouse to stream real-time updates directly to an interactive analytics dashboard.

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METHODOLOGY & DISCLOSURES APPENDIX

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Data Source: Raw meeting transcripts pulled programmatically from the central Federal Reserve Monetary Policy calendar archive (federalreserve.gov).

Algorithmic Architecture: Built using Python and the scikit-learn machine learning framework. Mathematical modeling utilizes scikit-learn's native TfidfVectorizer and cosine_similarity mathematical engines. Parameters tuned independently